From SMEFT weights to limits on Wilson coefficients

Goal: Estimate expected sensitivity on a single SMEFT operator

Build shapes & datacards

- Purpose: Convert generator-level weights into Combine-compatible histograms

1. Read mll, w_SM, w_SMEFT
2. Extract weights and build histograms: SM, LIN, QUAD
3. Compute integrals I_SM, I_LIN, I_QUAD (sum of bins)
4. Convert to unit-area shapes (so combine can use them)
5. Write shapes_<op>.root + datacard_mll_<op>_scaled.txt

Build all operators

- python3 make_shapes_all.py \
- --file trees/events.root \
- --ops 'clj1:2,clj3:4,ceu:6,ced:8,cje:10,clu:12,cld:14' \
- --outdir combine --channel ch1 --auto-mcstats

Datacard anatomy

- Purpose: Tell Combine how to combine SM and BSM pieces

- imax 1, jmax 2 → model structure
- shapes * ch1 shapes_ced.root ch1/$PROCESS → input link
- observation -1 → Asimov data
- process sm lin_ced quad_ced → components
- rate -1 1 1 → normalization control

Physics parameters

- N_pred = N_SM + r*|I_LIN|*lin_ced + r^2*I_QUAD*quad_ced

- rateParams:
- r_lin rateParam ch1 lin_ced (@0*|I_LIN|)   r
- r_quad rateParam ch1 quad_ced (@0*@0*I_QUAD) r

Physics parameters

- Header: declare structure of the model
- Shapes: point combine to the histograms
- Observation: define which data to use (Asimov)
- Process: define SM, linear, quadratic terms and their base rates
- rateParam: link EFT normalization to Wilson coefficients

Physics parameters

```
imax 1
jmax 2
kmax *

shapes * ch1 shapes_ced.root ch1/$PROCESS ch1/$PROCESS_$SYSTEMATIC

bin ch1
observation -1

# SM absolute; EFT are unit-area shapes (CED)
bin            ch1    ch1     ch1
process        sm           lin_ced       quad_ced
process        1            0             2
rate           -1           1             1

# Use Combine's built-in POI 'r' (scan r over +/-)
# Paste numbers already including any global scale:
r_lin    rateParam  ch1  lin_ced   (@0*340544426480.6179)      r
r_quad   rateParam  ch1  quad_ced  (@0*@0*993958622824947.8)     r
```

Make workspace

- text2workspace.py datacard_mll_ced_scaled.txt -o combine/workspace_ced.root

- Converts text datacard → RooFit workspace
- Want:
  - Grid scan of r to map delta(NLL)
  - Best fit and symmetric 68% / 95% CL intervals at r=0

Likelihood scan

- combine -M MultiDimFit workspace_ced.root \
- --algo grid --points 121 \
- --redefineSignalPOIs r \
- --setParameterRanges r=-0.1,0.1 \
- --setParameters r=0 -t -1

- Builds Asimov dataset at r=0 and scans ΔNLL(r)

Extract confidence intervals

- 68% CL (1 sigma):
- combine-M MultiDimFitcombine/workspace_ced.root--algo none\
  --redefineSignalPOIs r \

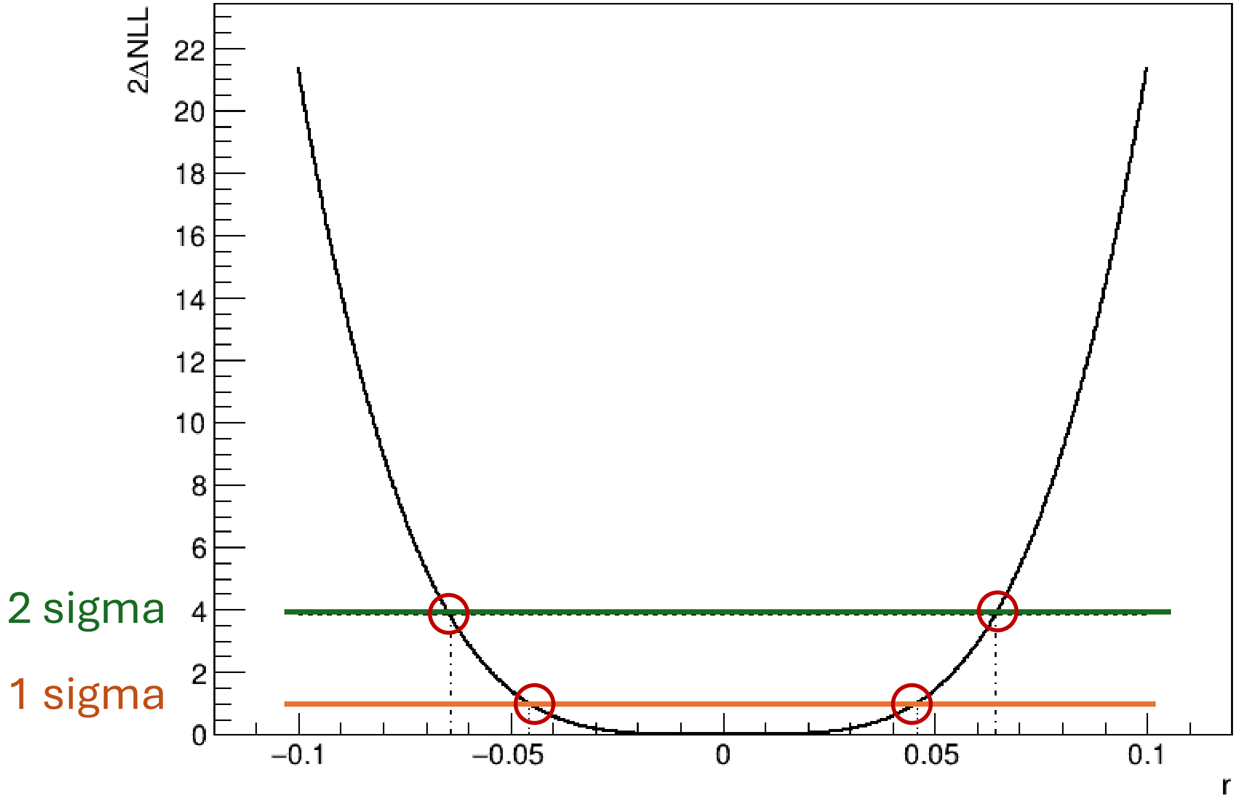
  --setParameterRanges r=-0.1,0.1 \

  --setParameters r=0 -t -1 --cl=0.68

- 95% CL (2 sigma): same with --cl=0.95

- Output: best-fit and ±intervals on r (expected sensitivity)

ΔNLL(r)-scan



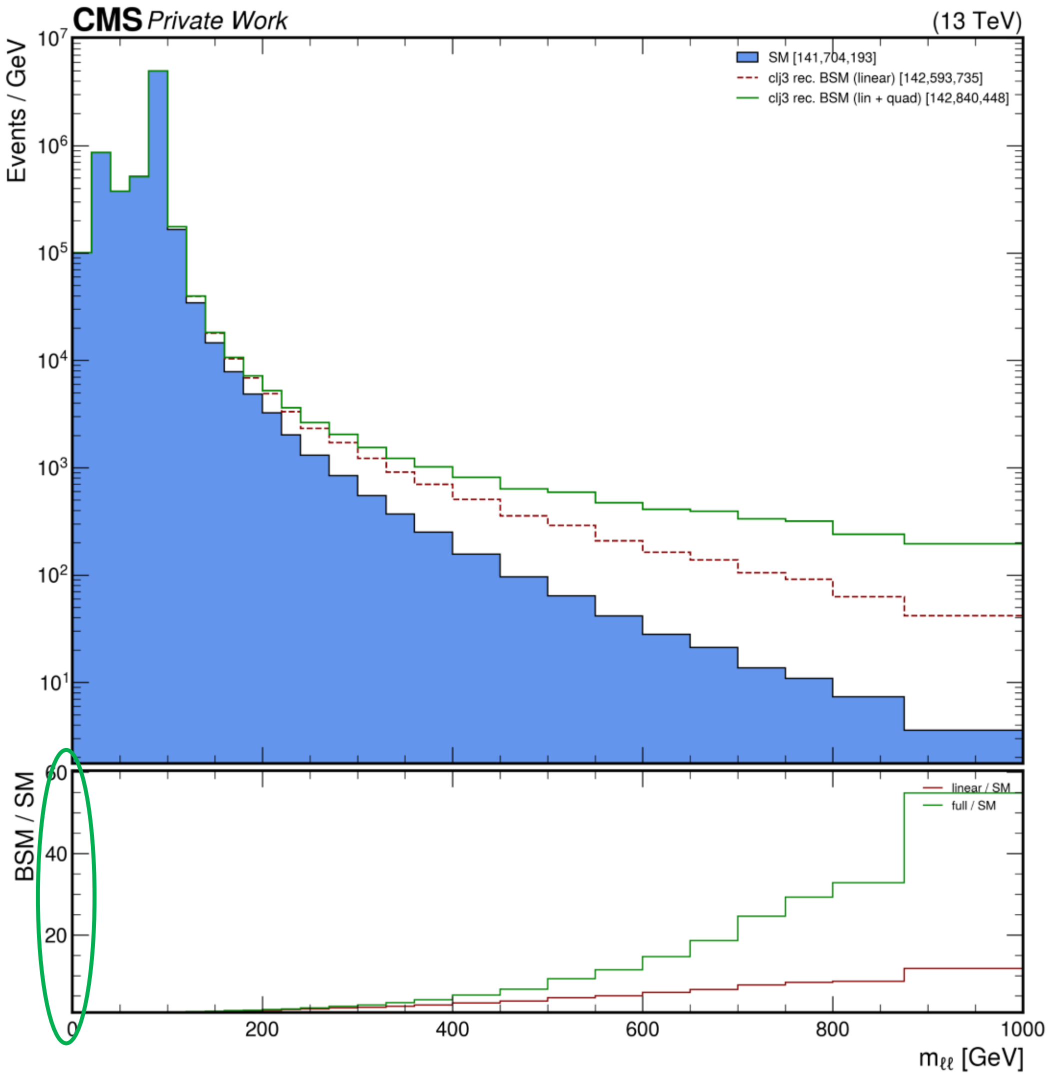
Expected sensitivity limits

operator	68% CL	95% CL
clj1	±0.01022	±0.01438
clj3	±0.02601	±0.03660
ceu	±0.00804	±0.01139
ced	±0.00294	±0.00426
cje	±0.00901	±0.01271
clu	±0.00311	±0.00451
cld	±0.00127	±0.00157

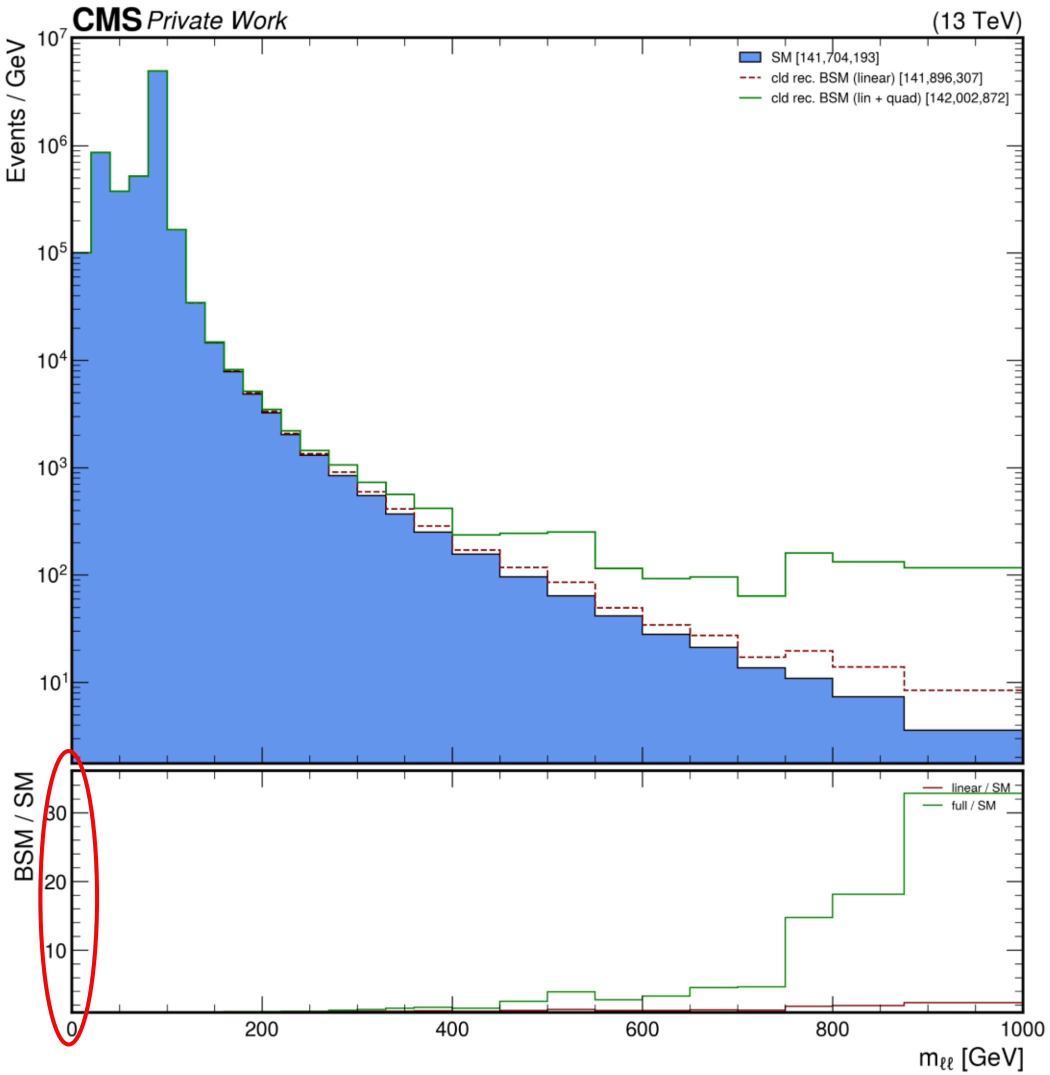
Low lim. -> high sens.

High lim. -> low sens.

High vs low sensitivity



clj3



cld

Key insights / takeaways

- Combine fits pre-binned templates, not events.
- r directly linked to SMEFT coefficient normalization.
- Validation with raw shapes is crucial to check physics behavior.
- Outputs: limits on r + sanity plots.

Goal & inputs

- Goal: Estimate expected sensitivity on a single SMEFT operator
- Input ROOT file: events.root:events
- - mll (dilepton mass)
- - EventWeight_SM & _SMEFT